

Bonus 2 - řešení!

$$\textcircled{1} \quad \text{a) } \left(\begin{array}{ccc|c} \textcircled{1} & 1 & 2 & 5 \\ 1 & 3 & 1 & 2 \\ 1 & 5 & -1 & 3 \end{array} \right) \begin{array}{l} \cdot (-1) \\ \leftarrow + \\ \leftarrow + \end{array} \sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & 5 \\ 0 & \textcircled{2} & -1 & -3 \\ 0 & 4 & -3 & -2 \end{array} \right) \begin{array}{l} \cdot (-2) \\ \leftarrow + \end{array} \sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & 5 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & -1 & 4 \end{array} \right)$$

$$\underline{\underline{x_3 = -4}}, \quad \begin{array}{l} 2x_2 + 4 = -3 \\ 2x_2 = -7 \Rightarrow \underline{\underline{x_2 = -\frac{7}{2}}} \end{array}, \quad \begin{array}{l} x_1 - \frac{7}{2} - 8 = 5 \\ x_1 = \frac{10 + 7 + 16}{2} = \underline{\underline{\frac{33}{2}}} \end{array}$$

$$\textcircled{b) } \begin{pmatrix} 1 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & 5 & -1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix}$$

$$\textcircled{2} \quad \begin{vmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{vmatrix} = (2-\lambda) \cdot (1-\lambda) - 6 = \lambda^2 - 3\lambda - 4 = (\lambda+1)(\lambda-4) = 0$$

$$\underline{\underline{\lambda_1 = -1}}, \quad \underline{\underline{\lambda_2 = 4}}$$

VLASTNÍ VEKTORY:

$$\lambda_1 = -1: \begin{pmatrix} 3 & 3 \\ 2 & 2 \end{pmatrix} \cdot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow u_1 + u_2 = 0 \Rightarrow \underline{\underline{\begin{pmatrix} 1 \\ -1 \end{pmatrix}}}$$

$$\lambda_2 = 4: \begin{pmatrix} -2 & 3 \\ 2 & -3 \end{pmatrix} \cdot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow -2u_1 + 3u_2 = 0 \Rightarrow \underline{\underline{\begin{pmatrix} 3 \\ 2 \end{pmatrix}}}$$